File 2: `OMNIGUARD_V2_ADVANCED_ROADMAP.md`

```
# OmniGuard AML (v2) — Model Fine-Tuning, End-to-End Product Automation, MLOps & Adva

## 1. Vision for OmniGuard v2
Building upon the verified, leakage-free architecture of **OmniGuard v1**, **OmniGuar

v2 addresses the core scaling boundaries of v1 (`n=338` training windows, `0.545` bas
1. **Multi-Tier Model Fine-Tuning:** Scaling tabular/graph ML with Bayesian hyperpara
2. **Closed-Loop Product Automation:** Connecting **ScenarioLab (Upstream)** directly
3. **Production MLOps & Continuous Governance:** Automated feature stores, drift dete
4. **Advanced Agentic AI Architecture:** Episodic memory, autonomous Corporate UBO &

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## 2. Pillar 1: Multi-Tier Model Fine-Tuning & Graph Deep Learning

### A. Tabular & Topological ML Optimization (Pushing LightGBM F1 from `0.545` $\righ
1. **Corpus Expansion & Class-Imbalance Engineering:**
   * Scale training ingestion from `338` windows to `100,000+` account-split windows
   * Replace static class weights with **Focal Loss** custom objectives and **SMOTE-N
2. **Automated Bayesian Hyperparameter Tuning (Optuna + MLflow):**
   * Run 200-trial Optuna sweeps optimizing **F2-Score** (weighting Recall $2\times$
   * Tune `num_leaves`, `max_depth`, `min_child_samples`, `colsample_bytree`, `subsam
3. **Dynamic Ensemble & Threshold Calibration:**
   * Replace the static `0.70 / 0.30` LightGBM-to-Isolation-Forest weighting with a *

### B. Deep Graph Neural Networks (From NetworkX Heuristics $\rightarrow$ Temporal Gr
* **Why:** 1-hop NetworkX metrics (`pagerank`, `betweenness`) plateau on multi-hop mo
* **v2 Implementation:**
  * Persist transaction graphs in **Neo4j / Memgraph** and train an inductive **Tempo
  * Generate 64-dimensional node and edge embeddings capturing temporal velocity and

### C. LLM & Embedding Fine-Tuning (LoRA / QLoRA + Direct Preference Optimization)
1. **Supervised Fine-Tuning (SFT) with QLoRA for Citation Fidelity (`0.667` $\rightar
   * Curate a gold-standard dataset of `5,000+` verified investigation runs from `aud
   * Fine-tune a compact open-weights model (**Llama-3.1-8B-Instruct** or **Nemotron-
   * **Result:** Eliminates reliance on congested 120B/552B free-tier endpoints, cutt
2. **DPO (Direct Preference Optimization) on Analyst Decisions:**
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   * Pair accepted agent runs (`audit.human_decisions.decision == agent_recommendatio
   * Run DPO alignment so the Investigator and Benign Challenger internalize institut
3. **Domain-Adapted Contrastive Embeddings (`pgvector` Upgrade):**
   * Fine-tune `bge-large-en-v1.5` using **Multiple Negatives Ranking Loss (MNRL)** o

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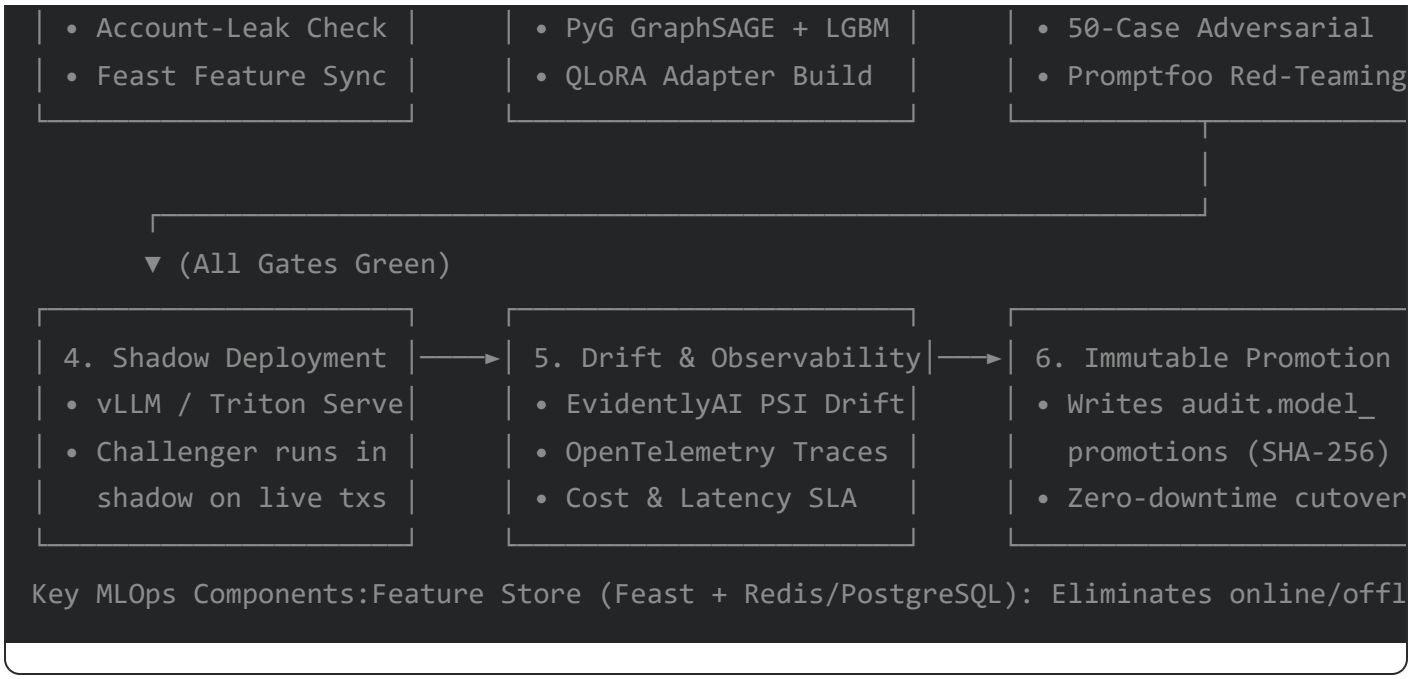
3. Pillar 2: Complete Product Automation (Closed-Loop Self-Healing AML)

In v1, ScenarioLab and the Investigation Pipeline run as separate workflows triggered

```text
┌──────────────────────────────────────────────────────────────────────────┐
│                  OMNIGUARD v2 CLOSED-LOOP AUTOMATION                     │
└──────────────────────────────────────────────────────────────────────────┘
       │
       ▼
[1. Regulatory Feed Watcher] ──► Scrapes FinCEN/FATF/RBI advisories -> Auto-drafts ne
       │
       ▼
[2. Autonomous ScenarioLab]  ──► Generates 100+ boundary mutations & lookalikes (SHA-
       │
       ▼
[3. Control Stress-Test]     ──► Runs current production LightGBM + SQL Rules against
       │
       ├──► If 100% Match ─────► Auto-generates Signed Compliance Evidence Pack (PDF
       │
       └──► If MISS / OVER-ALERT Detected:
                  │
                  ▼
         [4. Auto-Hardening Pipeline]
             • Converts missed synthetic scenarios into hard-positive / hard-negative
             • Triggers Optuna retraining job + Threshold Recalibration
             • Runs 50-case regression gate in MLflow
             • Promotes hardened model to staging & notifies Risk Owner for 1-click a
Additional Product Automation Modules in v2:Automated FinCEN SAR XML / PDF Generation
       │
       ▼
┌──────────────────────┐     ┌──────────────────────────┐     ┌──────────────────────────
│ 1. Data & Feature CI │────►│ 2. Continuous Training │────►│ 3. Multi-Gate Eval Suit
│ • Great Expectations │     │ • Optuna HPO Sweep     │     │ • Held-out F2 & PR-AUC
```


• Account-Leak Check
• Feast Feature Sync
• PyG GraphSAGE + LGBM
• QLoRA Adapter Build
• 50-Case Adversarial
• Promptfoo Red-Teaming
▼ (All Gates Green)
4. Shadow Deployment
• vLLM / Triton Serve
• Challenger runs in shadow on live txs
5. Drift & Observability
• EvidentlyAI PSI Drift
• OpenTelemetry Traces
• Cost & Latency SLA
6. Immutable Promotion
• Writes audit.model_promotions (SHA-256)
• Zero-downtime cutover
Key MLOps Components:Feature Store (Feast + Redis/PostgreSQL): Eliminates online/offl